MQTT – Current Sensor – Module

Summary:

Module for the ESP32 S3 based modular controller board.
Module is designed to run 3 x 4–20mA Sensors, and read them at a resolution of 15 bit.

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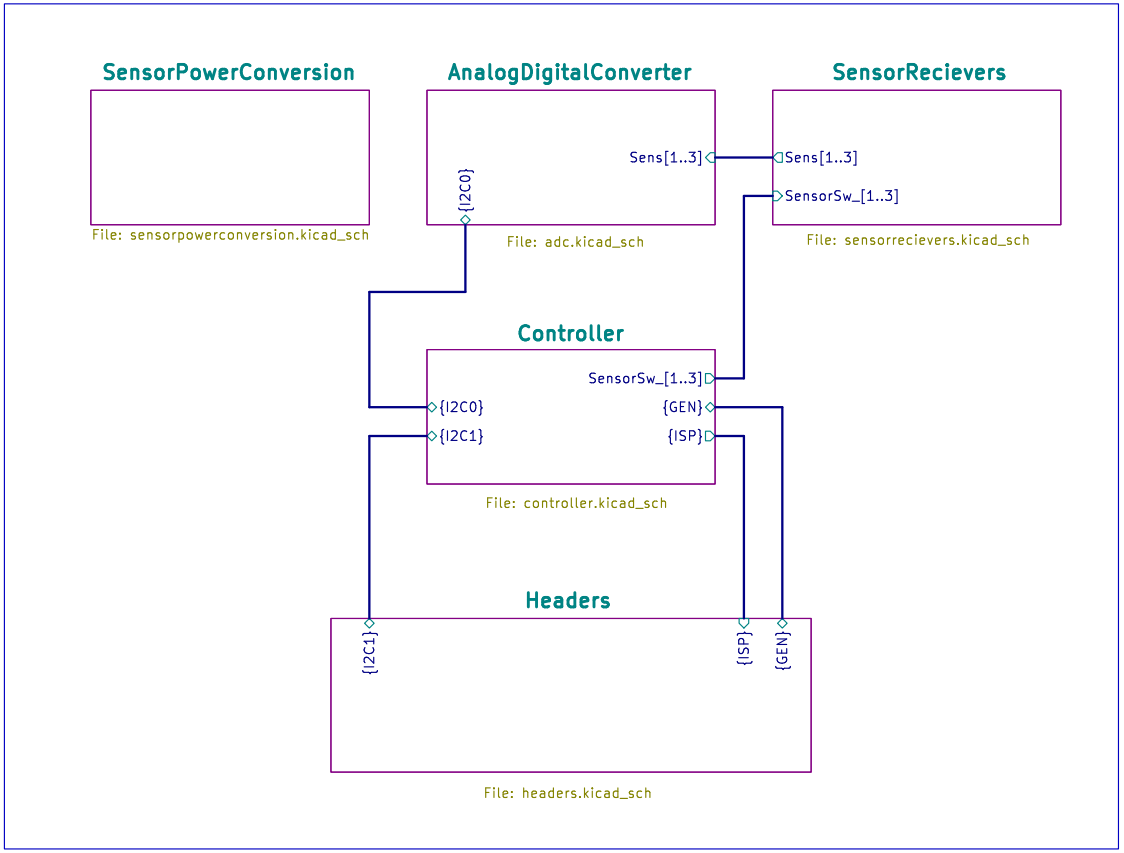
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Non BOM Items

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Additional BOM Items

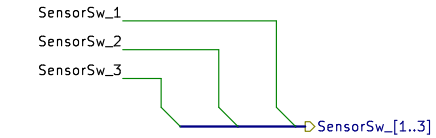
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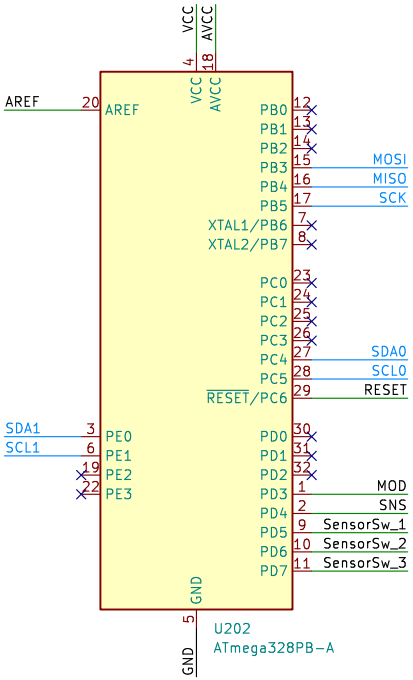


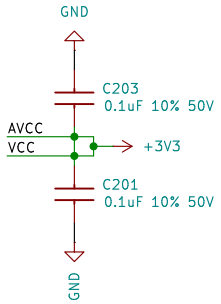
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File: MQTT–Module–CurrentSensors.kicad_sch

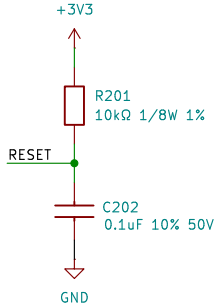
Title: MQTT – Current Sensor – Module

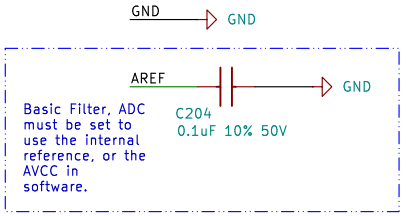
Size: A4	Date: 2026–10–09	**Rev: 1.1**
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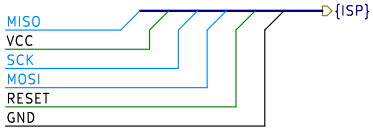


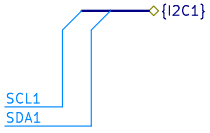


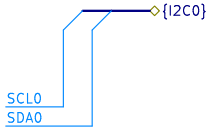


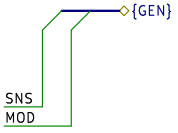










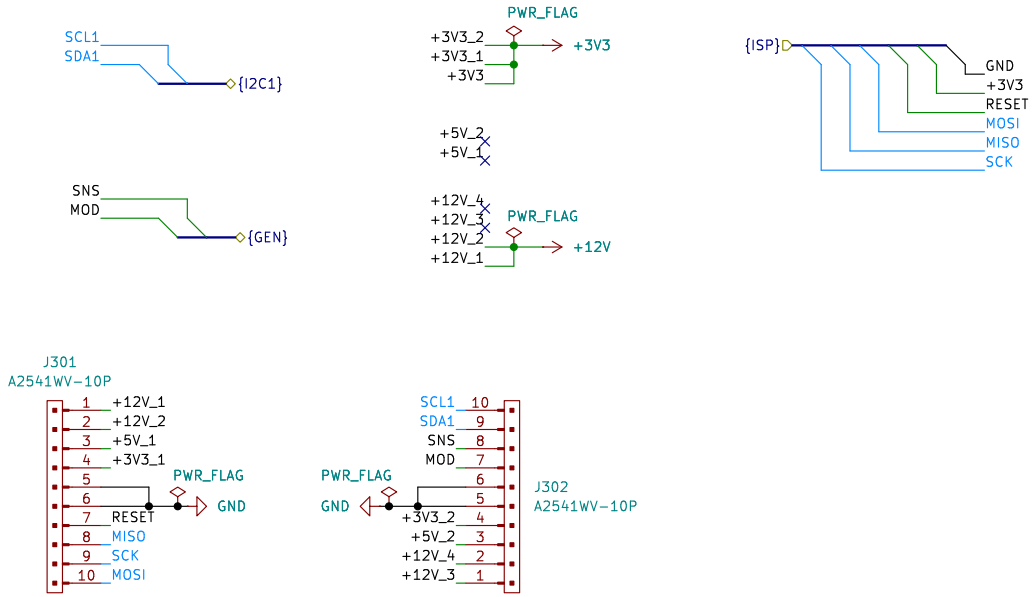


Summary:
– ATMega328PB–A Based control with dual dedicated SDA SCL.

Sheet: /Controller/
File: controller.kicad_sch

Title: Controller

Size: A4	Date: 2026-10-09	**Rev: 1.1**
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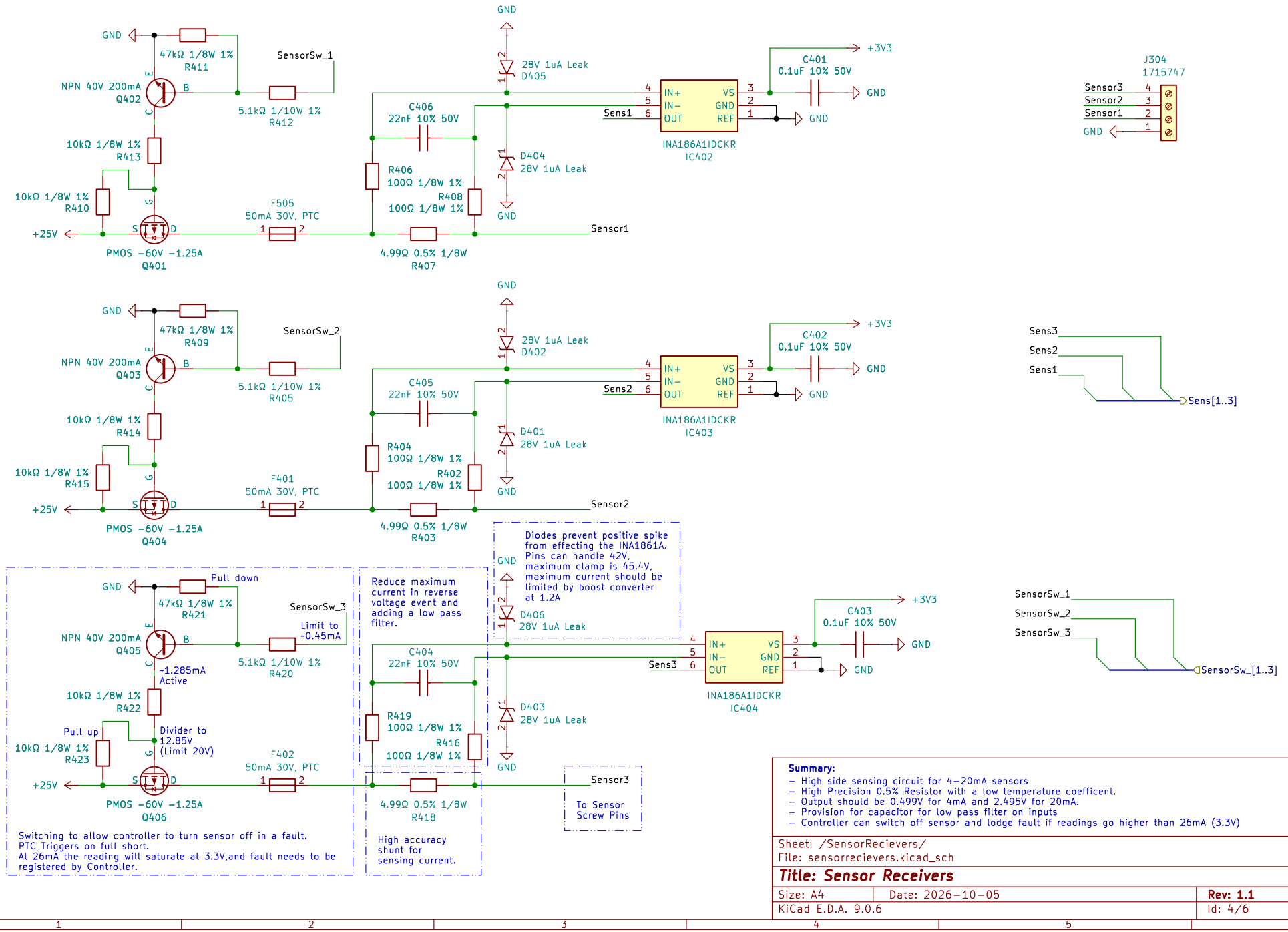


Summary:
- Headers to connect to main board. DNP as they will be hand soldered on.
- ISP Header for connection to MEGA ISP programmer.

Sheet: /Headers/
File: headers.kicad_sch

Title: Headers

Size: A4	Date: 2026-10-05	**Rev: 1.1**
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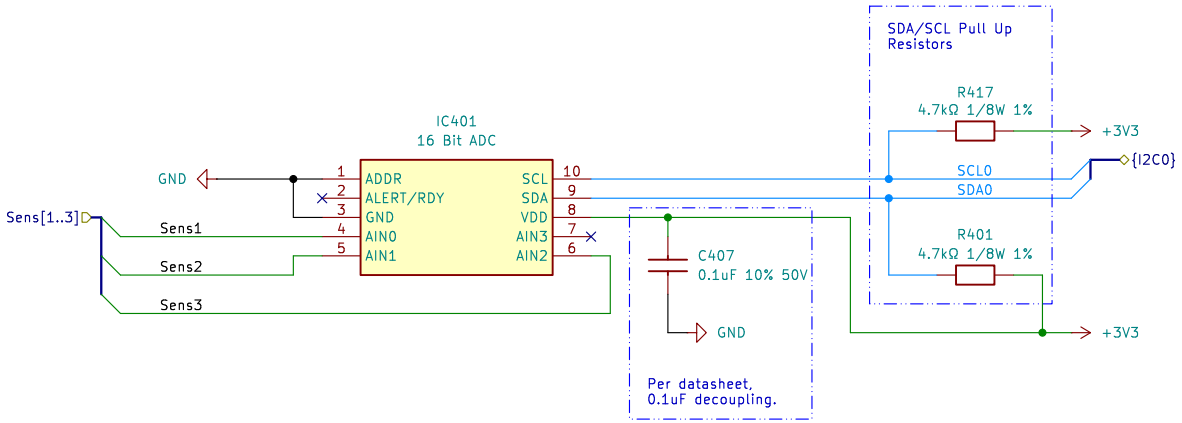
Summary:

- High side sensing circuit for 4–20mA sensors
- High Precision 0.5% Resistor with a low temperature coefficent.
- Output should be 0.499V for 4mA and 2.495V for 20mA.
- Provision for capacitor for low pass filter on inputs
- Controller can switch off sensor and lodge fault if readings go higher than 26mA (3.3V)

Sheet: /SensorRecievers/
File: sensorrecievers.kicad_sch

Title: Sensor Receivers

Size: A4	Date: 2026-10-05	**Rev: 1.1**
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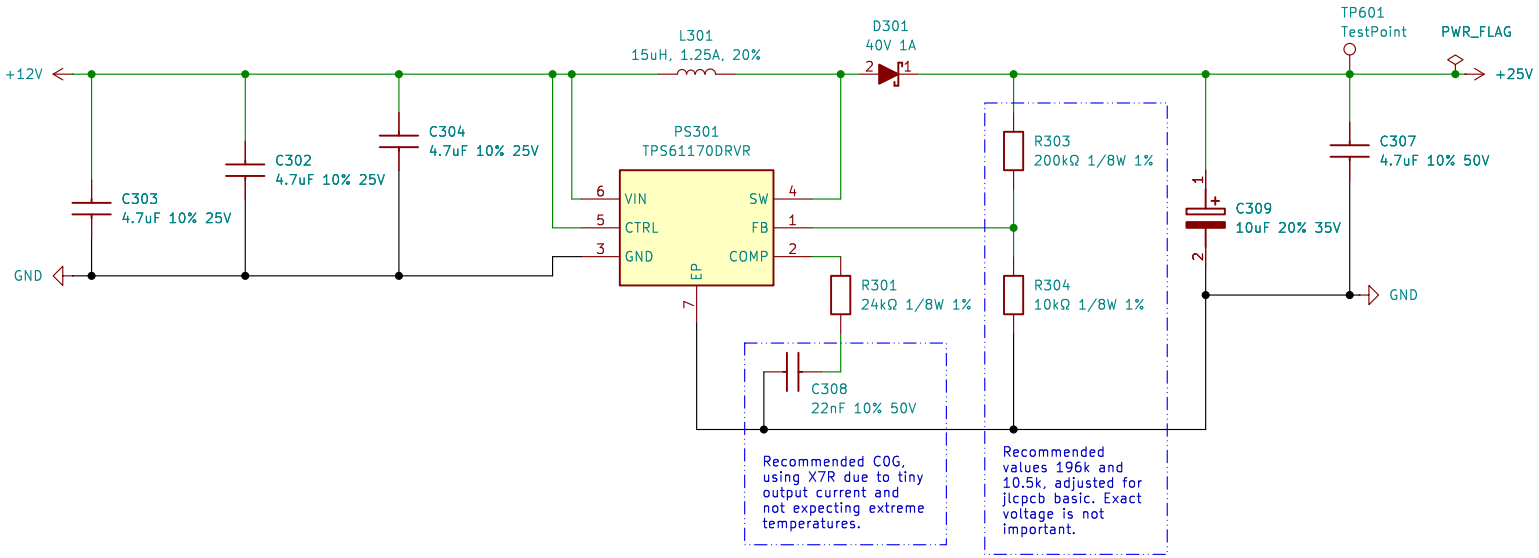


Summary:
– ADS1115 configured to use base address (0x48).

Sheet: /AnalogDigitalConverter/
File: adc.kicad_sch

Title: Analog Digital Converter

Size: A4	Date: 2026-10-09	**Rev: 1.1**
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Summary:
- Texas Instruments generated Boost Converter, 12V to ~25.7V
- Used a shielded inductor for minimal interference.

Sheet: /SensorPowerConversion/		
File: sensorpowerconversion.kicad_sch		
Title: Sensor Power Conversion		
Size: A4	Date: 2026-10-06	**Rev: 1.1**
KiCad E.D.A. 9.0.6		Id: 6/6